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Targeting up-regulated cIAP2 in SOX10-deficient drug tolerant melanoma

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Abstract

Drug tolerance and minimal residual disease (MRD) are likely to prelude acquired resistance to targeted therapy. Mechanisms that allow persister cells to survive in the presence of targeted therapy are being characterized but selective vulnerabilities for these subpopulations remain uncertain. We identified cellular inhibitor of apoptosis protein 2 (cIAP2) as being highly expressed in SOX10-deficient drug tolerant persister (DTP) melanoma cells. Here, we show that cIAP2 is sufficient to induce tolerance to MEK inhibitors, likely by decreasing the levels of cell death. Mechanistically, cIAP2 is upregulated at the transcript level in SOX10-deficient cells and the AP-1 complex protein, JUND, is required for its expression. Using a patient-derived xenograft model, we demonstrate that treatment with the cIAP1/2 inhibitor, birinapant, during the MRD phase delays the onset of resistance to BRAF inhibitor and MEK inhibitor combination therapy. Together, our data suggest that cIAP2 upregulation in SOX10-deficient subpopulations of

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Author Contributions: M.Q.G. designed research, performed experiments, analyzed the data, and wrote the manuscript. S.C., A.G.Y., N.W. and C.A.O. performed experiments and analyzed data. I.C. performed statistical analysis. K.T.F. and M.H. helped conceiving the *in vivo* experiment. X.X. analyzed IHC slides. A.E.A. provided his expertise to design some of the *in vitro* experiment related to cIAP2 regulation. C.C. conceived of the study, designed research, performed *in vivo* and *in vitro* experiments and wrote the manuscript.

Conflict of interest: A.E. Aplin reports ownership interest in patent number 9880150. Dr. Flaherty has served on the Board of Directors of Loxo Oncology, Clovis Oncology, Strata Oncology, Checkmate Pharmaceuticals, Kinnate Pharmaceuticals and Scorpion Therapeutics; Scientific Advisory Boards of PIC Therapeutics, Apricity, Oncoceutics, Fog Pharma, Tvardi, xCures, Monopteros, Vibliome, ALX Oncology, OMRx, Soley Therapeutics, and Quanta Therapeutics; and as consultant to Lilly, Novartis, Genentech, Takeda; and received research funds from Novartis and Sanofi. No potential conflicts of interest were disclosed by the other authors.

melanoma cells induces drug tolerance to MAPK targeting agents and provides a rationale to test a novel therapeutical approach to target MRD.

Keywords

cIAP2; minimal residual disease; drug tolerance; AP-1 complex; cIAP1/2 inhibitors

Introduction

Cutaneous melanoma is characterized by hyperactivation of the mitogen-activated protein kinase (MAPK) pathway, frequently by activating mutations in *BRAF*(1). Nearly all *BRAF* V600 mutant melanomas respond to BRAF inhibitor (BRAFi) plus MEK inhibitor (MEKi) treatment but acquired resistance inevitably develops (2). Targeted kinase therapies are less effective in wild-type (WT) *BRAF* melanomas, which are instead driven by either *NRAS* mutations, NF1 loss, or other alterations, that hyperactivate the MEK-ERK1/2 pathway (3). To date, there are no effective targeted therapy options for WT *BRAF* melanoma patients. Immune checkpoint inhibitors (ICi), including anti-CTLA-4 (ipilimumab), anti-PD-1 (nivolumab and pembrolizumab) and anti-LAG-3 (relatlimab) antibodies, significantly improve overall survival for Stage III and IV melanoma patients (4, 5, 6, 7). ICi are efficacious in 14-60% of melanoma patients but immune-related adverse effects are common (4, 5, 6). If WT *BRAF* melanoma patients progress on ICi, there are limited second-line options.

MRD is defined as a small subpopulation of drug tolerant persister (DTP) cells that remain viable upon drug exposure, whereas the rest of the cell population is rapidly killed (8). In melanoma, multiple cellular states are associated with MRD and exist before the development of acquired resistance to either BRAFi plus MEKi or ICi (8, 9, 10). MRD cell states are characterized by the activation of different transcriptional programs including the MITF/pigmented state; the starvation-like transcriptional program; a Neural Crest-like Cell (NCSC) state expressing NGFR; and the invasive/mesenchymal cell state characterized by overexpression of AXL, loss of SOX10 and MITF, and activation of the AP-1 complex transcription factor (8, 11). Despite these studies, mechanisms that promote DTP cell survival and could serve as targetable vulnerabilities in MRD subpopulations remain poorly defined.

We showed upregulation of cellular Inhibitor of apoptosis protein 2 (cIAP2) in SOX10-deficient cells. IAPs, including XIAP (X-linked IAP), cIAP1, cIAP2, and survivin, are a family of structurally-related proteins and their levels correlate with treatment response and prognosis in many cancers (12). These proteins govern numerous cellular processes, such as activation of the nuclear factor- κ B (NF- κ B) pathway and apoptosis through their E3 ligase activity (12). However, the mechanisms driving cIAP2 upregulation in melanoma DTP cells are not known. Additionally, several cIAP1/2 inhibitors (SMAC mimetics) have entered clinical cancer trials, including GDC-0512, LCL16, SM-406/AT-406, birinapant and ASTX660 (13). SMAC mimetics may have limited effectiveness as individual agents (13); thus, combination therapies should be evaluated in preclinical models (13).

Here, we examined cIAP2 (encoded by the *BIRC3* gene) upregulation as a mechanism of drug tolerance across genotypes in melanoma. We show that both *BRAF*^{WT} and *BRAF* mutant drug tolerant melanoma cells express high levels of cIAP2. cIAP2 overexpression is sufficient to induce tolerance to MEKi, likely by reducing MEKi-induced cell death. Mechanistically, cIAP2 is up-regulated in SOX10-deficient cells at the transcript level by the AP-1 protein complex component JUND. cIAP2 expressing cells can be targeted with the IAP inhibitor, birinapant, through induction of cell death. In patient-derived xenograft (PDX) *in vivo* studies, adding birinapant to BRAFi plus MEKi, following regression to the MRD phase, delays the onset of resistance. Together, our findings provide a novel therapeutical approach to delay the onset of targeted therapy resistance by reducing/eliminating DTP cells during the MRD stage.

MATERIALS AND METHODS

Cell culture:

MeWo parental (donated by Dr. Barbara Bedogni in 2014), MeWo plentipuro/TO/V5/luciferase-m-Cherry (MeWo-Luc-m-Cherry), A375 parental (purchased from ATCC in 2005), MeWo-LacZ or MeWo-cIAP2 overexpressing cells, MeWo crSOX10 #4.11-LacZ, or MeWo crSOX10 #4.11-HA-SOX10, and previously generated A375 and MeWo crSOX10 knockout cells (14) were cultured in DMEM (Gibco, Billings, MT, #11966-025) with 10% FBS (Gemini, New York, NY, #900-108) and 1% penicillin/streptomycin (Corning, Corning, NY, #30-002-CI). A375 BRAFi plus MEKi tolerant cells (CRT34, CRT35, previously referred to as CRT14 and CRT15) (15) were cultured in the presence of PLX4720 (1 μ M) (16, 17) and PD0325901 (18) (35 nM). MeWo MEKi Tolerant Cells (MTC) were generated from MeWo-Luc-m-Cherry cells (14) by using increasing concentrations of trametinib until 50 nM, and then were permanently cultured in the presence of 50 nM trametinib. *BRAF* and *NRAS* mutation status in cell lines was validated by Sanger sequencing. Cells were tested for mycoplasma contamination every month with MycoScope Kit (Genlantis, San Diego, CA, #MY01100). Short-tandem repeat analysis was completed for MeWo parental, MeWo crSOX10 knockout cells (April 2019), MeWo-Luc-m-Cherry, MeWo MTC (October 2021), A375 parental, A375 crSOX10 knockout cells, and CRT cells (October 2020). All cell lines matched a known profile. Cells were kept in culture for a maximum of 10 weeks. Cells were cultured at 37°C with 5% CO₂ in a humidified chamber.

Inhibitors:

PD0325901 (PD'901) (18) and trametinib were purchased from Selleck Chemicals, Houston, TX. Birinapant (19) was provided by the National Cancer Institute (NCI) Division of Cancer Treatments and Diagnosis (DCTD). PLX4720 (16, 17) and PD'901 (17, 18) were provided by Dr. Gideon Bollag (Plexxikon Inc., Berkeley, CA).

Annexin V / Propidium Iodide (PI) Staining:

Cells were plated in 6-well plates and treated as indicated in the figure. After 48 hrs, cells were harvested, washed twice with 1X PBS, and incubated in Annexin V binding buffer with Annexin V-APC (BD Pharmingen, Franklin Lakes, NJ, #550475, 1:100) and PI (Thermo Fischer Scientific, Waltham, MA, #P1304MP, 1mg/mL) for 15 min. at room temperature in

the dark. Samples were then resuspended in binding buffer and analyzed using a BD Celesta or BD FACS Symphony A5 flow cytometer.

Analysis of publicly available chromatin immunoprecipitation-sequencing (ChIP-seq) data:

AP-1 protein complex ChIP-seq data from the ENCODE consortium (20, 21) was visualized using the UCSC Genome Browser in hg38. Visualized track sets were for FOSL2 (*ENCFF808RWZ*, *ENCFF054ESU*), JUN (*ENCFF127HJG*, *ENCFF312GEN*, *ENCFF032UMW*, *ENCFF167WUZ*, *ENCFF394CEC*, *ENCFF672LKE*, *ENCFF881AVX*, *ENCFF907UNK*), JUND (*ENCFF587VEY*, *ENCFF873DJD*, *ENCFF443HNU*, *ENCFF646IUA*, *ENCFF998KDQ*, *ENCFF430PEI*, *ENCFF539GRW*, *ENCFF213EYD*, *ENCFF229COM*, *ENCFF420PED*, *ENCFF569ZCY*, *ENCFF187QQB*, *ENCFF246HKM*, *ENCFF946YUA*), and MAFK (*ENCFF813WJW*, *ENCFF186AWV*, *ENCFF712RIS*, *ENCFF328IZQ*, *ENCFF171OJF*, *ENCFF770TZL*, *ENCFF351VGZ*, *ENCFF893SCL*, *ENCFF873SVI*). The putative *BIRC3* promoter was identified using the ENCODE Registry of candidate cis-Regulatory Elements (cCREs) database of candidate promoter-like signatures. SOX9 ChIP-seq data from human colorectal cancer cells, HT-29, published by Shi et al. (22) was visualized using the UCSC Genome Browser in hg19. SOX10 ChIP-seq data from 501Mel melanoma cells published by Laurette et al. (23), and from MNT-1 melanoma cells published by Eskiocak et al. (24), was visualized using the UCSC Genome Browser in hg38.

Western blot analysis:

Lysate preparation and western blot analysis was performed as previously described (14). Primary antibodies for cIAP2 (#3130, 1:500, RRID: AB_10693298), HSP90 (#4877, 1:3000, RRID: AB_2233307), V5 (#13202, 1:1000, RRID: AB_2687461), pRB (807/811) (#9308, 1:1000, RRID: AB_331472), RB (#9302, 1:1000), pERK1/2 (#9102, 1:1000, RRID: AB_330744), cleaved PARP (#9541, 1:1000, RRID: AB_331426), PARP (#9542, 1:1000, RRID: AB_2160739), cIAP1 (#7065, 1:1000, RRID: AB_10890862), SOX10 (#89356, 1:1000, RRID: AB_2792980), p21^{Cip1} (#2947, 1:1000, RRID: AB_823586), WNT5 (#2530, 1:1000, RRID: AB_2215595), NFκB 105/50 (#3035, 1:1000, RRID: AB_330564) SOX9 (#82630, 1:1000, RRID: AB_2665492), and cleaved caspase-3 Asp175 (#9664S, 1:500, RRID: AB_2070042) were purchased from Cell Signaling Technology, Danvers, MA. FN1 (#ab45688, 1:1000, RRID: AB_732380) and GSDME (#ab215191, 1:500, RRID: AB_2737000) antibodies were purchased from Abcam, Cambridge, UK. Actin (#A2066, 1:2000, RRID: AB_476693) antibody was purchased from Sigma-Aldrich Co, St. Louis, MO. JUND (#sc-74, 1:1000, RRID: AB_2130177), FOSL2 (#sc-604, 1:1000, RRID: AB_2107084), p65 NFκB (#sc-109, 1:1000, RRID: AB_632039), and Cyclin A (#sc-271682, 1:1000, RRID: AB_10709300) antibodies were purchased from Santa Cruz Biotechnology, Dallas, TX. Secondary goat anti-mouse IgG (#401215, 1:4000, RRID: AB_10682749) and goat anti-rabbit IgG (#401315, 1:4000 or 1:1000, RRID: AB_437787) antibodies were purchased from Sigma-Aldrich Co, St. Louis, MO.

Crystal Violet:

MeWo-Luc-m-cherry or MeWo MTC were plated in 6-well plates (1×10^5 per well) and treated as indicated in the figure. At the end of the experiment, cells were washed

twice with PBS and stained with 0.2% crystal violet in 10% buffered formalin. Plates were subsequently washed with water and air-dried before being imaged on a Nikon A1R confocal microscope at 4x magnification using NIS-Elements AR Analysis 5.11.03 software (Nikon, Japan), and quantification was determined using ImageJ. Cell viability was analyzed by crystal violet staining.

Cell growth assay (IncuCyte):

Cell growth analysis via IncuCyte was performed, as previously described (14). Forty-nine pictures per well were taken every 2 hrs. Shown in the figure is a representation of percent plate coverage every 24 hrs.

Lentiviral constructs:

To generate HA-BIRC3 and LacZ cDNA lentiviruses, pLentiHygro/TO/V5-DEST was used. Human HA-tagged BIRC3 and control LacZ were cloned into the entry vector pENTR/D-TOPO (Invitrogen, Waltham, MA) and recombined into pLentiHygro/TO/V5-DEST. To generate HA-SOX10 and LacZ cDNA lentiviruses, pLentiPuro/TO/V5-DEST was used. Human HA-tagged SOX10 (25) and LacZ control were cloned into pENTR/D-TOPO and recombined into pLentiPuro/TO/V5-DEST. Lentivirus was created by packaging the resulting destination constructs into 293T cells, as previously described (26).

Generation of LacZ, cIAP2, and SOX10 overexpressing cells:

To generate MeWo-cIAP2 and MeWo-LacZ cells, MeWo parental cells were transduced with pLentiHygro/TO/V5/GW/LacZ or pLentiHygro/TO/V5/GW/HA-BIRC3 lentivirus and selected for using hygromycin (InvivoGen, San Diego, CA, #ant-hg) at 200 µg/ml. To generate MeWo crSOX10 #4.11-LacZ and MeWo crSOX10 #4.11-HA-SOX10, MeWo crSOX10 #4.11 cells were transduced with pLentiPuro/TO/V5/GW/LacZ or pLentiPuro/TO/V5/GW/HA-SOX10 lentivirus and selected for using puromycin (Gibco, Billings, MT, #A11138-03) at 1 µg/ml.

Quantitative Real Time Polymerase Chain Reaction (qRT-PCR):

RNA was extracted using the QIAGEN RNeasy Mini Kit. cDNA was synthesized from RNA using the Bio-Rad iScript cDNA Synthesis Kit. Quantitative RT-PCR was performed using SsoAdvanced Universal SYBR Green Supermix (Bio-Rad, Hercules, CA) and primers manufactured by IDT, Newark, NJ. *BIRC3* expression was calculated after normalization to the geometric mean Ct values of the reference gene GAPDH. Primer sequences for qRT-PCR are as follows: GAPDH (Forward): 5'- GTC TCC TCT GAC TTC AAC AGC G-3', GAPDH (Reverse): 5'- ACC ACC CTG TTG CTG TAG CCA A-3', BIRC3 (Forward): 5'- GCT GTG ATG GTG GAC TCA GG-3', BIRC3 (Reverse): 5'- ACT GGC TTG AAC TTG ACG GA-3'.

RNA-Seq and Volcano Plot analysis:

Raw FASTQ RNA sequencing reads for MeWo and A375 parental and crSOX10 cells were obtained from a publicly available dataset under the NCBI Bioproject numbers PRJNA701949 and PRJNA748713 (14). A list of transcription factors that are predicted

to bind to the *BIRC3* promoter was generated using Harmonizome, and transcription factors that were significantly differentially regulated across all of the A375 and MeWo crSOX10 knockout models, as compared to parental cells, were included in the figure.

Small interfering RNA (siRNA) transfection:

A375 crSOX10 knockout cells were transfected with either siGenome non-targeting control (Dharmacon, Lafayette, CO, #CTM-602672/APLAB-000061), siRNA targeting JUND (Dharmacon, Lafayette, CO, #D-003900-09, #D-003900-08) or siRNA targeting FOSL2 (Dharmacon, Lafayette, CO, #D-004110-01) at a final concentration of 25 nM using Lipofectamine RNAiMAX (Invitrogen, San Diego, CA, #56532). Sequences are as follows: JUND#9, GAAACACCCUUCUACGGCG; JUND#8, CGCUCAAGGACGAGCCACA; FOSL2#1, GGCCCAGUGUGCAAGAUUA; Non-targeting siGenome control, UAGCGACUAAACACAUCAAUU. MeWo and A375 parental and crSOX10 knockout cells were transfected with either on-target control siRNA (Dharmacon, Lafayette, CO, #D-001810-01) or siRNA targeting cIAP1 (Dharmacon, Lafayette, CO, #J-004390-12), and MeWo MTC, A375 CRT34 cells, and A375 CRT35 cells were transfected with either siGenome non-targeting control (Dharmacon, Lafayette, CO, #CTM-602672/APLAB-000061), siRNA targeting NFκB p50 (Dharmacon, Lafayette, CO, #D-003520-01), or siRNA targeting NFκB p65 (Cell Signaling, Danvers, MA, #6261) at a final concentration of 25 nM using Lipofectamine RNAiMAX (Invitrogen, San Diego, CA). Sequences are as follows: cIAP1, UCGCAAUGAUGAUGUCAA; NFκB p50, GCAGGUUUUGACAUUUUA; NFκB p65, the sequence is not publicly available; On-target control, UGGUUUACAUGUCGACUAAUU; Non-targeting siGenome control, UAGCGACUAAACACAUCAAUU.

Immunohistochemistry:

Naïve TJUMEL 41 sample was formalin-fixed and paraffin-embedded (FFPE), sectioned, and stained with SOX10 antibody (Abcam, Cambridge, UK #ab227680, 1:200). Selected areas of SOX10-deficient cells were analyzed by board-certified pathologist Dr. Xu (University of Pennsylvania) and identified as cancer cells. TJUMEL41 PDX treated as indicated in Fig. 4 and collected at the end of the experiment were FFPE, sectioned, and stained with H&E.

Analysis of publicly available Assay for Transposase-Accessible Chromatin-sequencing (ATAC-seq) data:

ATAC-seq data following SOX10 knockdown (0, 24, 48, and 72 hrs) for the intermediate MM057 primary melanoma culture published by Wouters et al. (11), was visualized using the UCSC Genome Browser in hg38.

Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR):

To generate CRISPR SOX9 knockout cells, A375 CRT34 and CRT35 cells were co-transfected with Edit-R Cas9 plasmid containing a puromycin-resistance marker (Horizon Discovery, Waterbeach, UK, U-005100-120), two different crRNAs for SOX9 (guide#01, Horizon Discovery, Waterbeach, UK, CM-021507-01-0002, target sequence

AGCCTTGCCCGGCTGCACGT, and guide#02, Horizon Discovery, Waterbeach, UK, CM-021507-02-0002, target sequence GATGCCGTAGCTGCCCCGTGT), and tracrRNA using DharmaFECT Duo (Horizon Discovery, Waterbeach, UK). After 48 hrs, puromycin was added to the medium and cells were selected for growth. Individual clones were expanded and tested for SOX9 knockout by Western blotting.

Single-cell RNA-sequencing (scRNA-seq) data analysis:

The scRNA-seq data for 10 cutaneous melanoma cell lines (11) was obtained from http://scope.aertslab.org/#/Wouters_Human_Melanoma. The website was used to generate tSNE plots showing scRNA-seq expression data for *SOX10*, *SOX9* and *BIRC3*.

3D spheroid:

A375 CRT cells were plated at 5×10^3 cells/well in an agarose-coated 96-well plate and left to form spheroids over 3 days. Spheroids were then collected using a transfer pipette, resuspended in a collagen mixture (rat tail Collagen Type I with Reconstitution Buffer, Ham's F-12 Nutrient Mix, and 5% FBS), and plated in a collagen-coated 24-well plate. After setting for 30 min. at 37°C, 1ml of normal culture medium was added to each well. After resting for 24 hrs, spheroids were treated with BRAFi plus MEKi and/or birinapant for 4 days; treatment was renewed every 48 hrs. Spheroids were then stained with Calcein AM (Invitrogen, San Diego, CA, #C1430) and Propidium Iodide (Invitrogen, San Diego, CA, #P3566) diluted in PBS for 30 mins. Spheroids were imaged on the Nikon A1R Microscope using a 4x objective lens and NIS-Elements software (Nikon, Japan).

In vivo studies:

Animal experiments were performed at a Thomas Jefferson University facility that is accredited by the Association for the Assessment and Accreditation of Laboratory Animal Care. The Institutional Animal Care and Use Committee at Thomas Jefferson University approved these studies (Protocol #: 01052). All animals were provided with food and water *ad libitum* and housed in cages (with a maximum of 5 mice/cage) in a temperature and humidity-controlled environment. Animals were maintained in housing conditions that allow for normal species behavior to minimize the development of abnormal behaviors, and have access to humane and veterinary care. *BRAF* mutant patient-derived xenograft (PDX) TJUMEL 41 was obtained from a BRAFi plus MEKi naïve patient sample obtained from a fifty-one-year-old male. PDX were generated in NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ (NSG) mice bred in house (RRID: BCBC_4142). When tumors were palpable, mice were treated with BRAFi plus MEKi (PLX4720 200 PPM + PD'901 7 PPM) until MRD was reached (4 weeks). Mice were then randomly sorted in 4 cohorts: i) mice were treated with BRAFi plus MEKi chow (PLX4720 200 PPM + PD'901 7 PPM); ii) In biweekly cycles: mice were treated for 1 week with BRAFi plus MEKi chow (PLX4720 200 PPM + PD'901 7 PPM), the following week control chow was administered and mice were injected intraperitoneally, twice per week, with 100 μ L (for female mice) or 150 μ L (for male mice) birinapant solution (formulated at 3 mg/ml in 5% DMSO and 15% captisol, CyDex Pharmaceuticals, San Diego, CA, aqueous solution) iii) In weekly cycles: mice were treated for four days with BRAFi plus MEKi chow (PLX4720 200 PPM + PD'901 7 PPM), on the 4th day food was switched to control chow, on the 5th day mice were treated with birinapant formulated, as

described above, and on day 7th food was switched to BRAFi plus MEKi chow (PLX4720 200 PPM + PD'901 7 PPM). iv) Mice were treated continuously with BRAFi plus MEKi (PLX4720 200 PPM + PD'901 7 PPM) and injected intraperitoneally, twice per week, with 100 μ L with birinapant formulated as described above. To prevent weight loss, after two weeks of treatment, the mouse diet was supplemented with *ClearH₂O* DietGel® Recovery (Bio-Serv, San Diego, CA) and additional supplement (Bio-Serv, San Diego, CA, #F3580) one day per week. Three mice were excluded from the study due to severe weight loss. Specifically, 9 mice were enrolled in the BRAFi plus MEKi and birinapant continuous arm. However, due to severe weight loss prompted by causes that we were not able to identify in the present study, two mice were sacrificed and removed from the study. For similar reasons, 7 mice of the original 8 are shown for the BRAFi plus MEKi arm. Digital caliper measurements of the tumors were taken twice per week, and tumor volumes were calculated using the formula: volume = (length $\times$ width²) $\times$ 0.52. Mouse weights were monitored once a week.

Statistical analysis:

For the IncuCyte experiment comparing MeWo-LacZ and MeWo-cIAP2 overexpressing cells, statistics were performed. For each replicate within treatment and dose, the confluency at each time was first normalized by the confluency at time zero (0). The resulting ratios were linearly interpolated between time points 1-14 and the area under the resulting curves (AUC) was computed on the time interval for plate. Log-transformed plate-specific AUCs were analyzed in a two-way ANOVA model with the nominal effect of dose, cell line (MeWo-LacZ vs MeWo-cIAP2 overexpressing cells) and their interaction. The model was used to estimate the mean difference in log AUCs between MeWo-LacZ and MeWo-cIAP2 cells at each dose level of trametinib. For qRT-PCR analysis, normalized *BIRC3* expression for each replicate in each cell line for each experiment (date) was computed as $2^{\Delta(\text{meanCq}(\text{GAPDH}) - \text{Cq}(\text{BIRC3}))}$, where meanCq(GAPDH) is the mean Cq of 3 replicates in each cell line-by-experiment combination. Then the log-transformed normalized *BIRC3* expression for all *BIRC3* replicates were modeled in a general linear model with the effects of cell line and experiment. Separate models were fitted to data for MeWo and its clones and to data for A375 and its clones. The fitted models were used to compare the difference in normalized *BIRC3* expression between each clone and parental cell line. For Annexin/PI staining, percent staining was analyzed in a two-way ANOVA model. For the volcano plot RNA-seq analysis, a list of transcription factors that are predicted to bind to the *BIRC3* promoter was generated using Harmonizome. For each gene, the normalized sequence counts in each replicate were log base 2 transformed and modelled in one-way ANOVA model with six groups representing the 2 clones and parental cells from MeWo and A375 cell lines. Each fitted model was used to test the 4-dimensional contrast corresponding to simultaneously testing the differences in mean log2-transformed normalized sequence counts between (i) MeWo crSOX10 #2.1 vs. parental MeWo; (ii) MeWo crSOX10 #4.11 vs. parental MeWo; (iii) A375 crSOX10 #2.18 vs. parental A375; (iv) A375 crSOX10 #4.21 vs. parental A375. The *p*-values from all models (for all considered genes) were subsequently adjusted for multiple testing using the Hochberg method (27) controlling for the family-wise error rate at the 0.05 level. Among significantly differentially expressed genes (with adjusted *p*-values <0.05), the genes were selected as upregulated if all mean differences (i)-(iv) were

positive and as downregulated if all mean differences (i)-(iv) were negative. For spheroid fluorescence intensity, statistical analysis was performed using Student's two-sample t-test assuming unequal variance. For the *in vivo* experiment comparing survival for differing treatment arms, statistics were performed as follows: the time to sacrifice for each animal was computed as the number of days for the tumor volume to exceed 1000 mm³. Log transformed days to sacrifice were analyzed in one-way ANOVA model with treatment group as a predictor. Post-hoc paired comparisons were performed with Tukey family-wise adjustment for multiple comparisons. Corresponding survival curves were analyzed using a Logrank (Mantel-Cox) test. The analysis was performed in R (The R Foundation for Statistical Computing <http://www.R-project.org>). Cyclin A and phosphoRB levels were quantified using Image Lab in MeWo-Luc-m-Cherry and MeWo MTC following treatment with trametinib. For each replicate within a treatment and cell line, the expression of cyclin A and phosphoRB was normalized to HSP90 expression. The log-transformed normalized expression of cyclin A and phosphoRB was analyzed in separate two-way ANOVA models with the indicator of trametinib treatment (vs. DMSO), cell line (MeWo-Luc-m-Cherry or MeWo MTC) and their interaction. The effect of replicate (experiment day) was considered but not included into the final models due to non-significance. The models were used to estimate the effect of trametinib treatment separately in each cell line and compare the ratios that quantify the difference in effect of trametinib treatment between cell lines using model-based contrast. The analysis was performed in R (The R Foundation for Statistical Computing <http://www.R-project.org>). For the crystal violet experiment, experiments were performed in triplicate and statistical significance was determined using a two-way ANOVA test. For the incucyte experiment shown in the supplemental figures, p-values were calculated as area under the curve using two-sided t-test.

Data Availability Statement:

Raw FASTQ RNA sequencing reads for MeWo and A375 parental and crSOX10 cells were obtained from a publicly available dataset under the NCBI Bioproject numbers PRJNA701949 and PRJNA748713 (14). Harmonizome was used to generate a list of transcription factors that are predicted to bind to the *BIRC3* promoter. All the other data supporting this study are available within this Article, Supplementary Information or from the corresponding authors upon reasonable requests.

Results

cIAP2 is sufficient to induce tolerance to MEKi

While drug tolerant cells have been characterized by transcriptomic states, the mechanisms that allow persister cells to survive in the presence of treatment remain poorly defined. cIAP2 is up-regulated in SOX10-deficient DTP melanoma cells (14), but the functional role of cIAP2 in mediating tolerance to MAPK targeting agents remains unknown. We tested whether cIAP2 was sufficient to mediate drug tolerance to MAPK inhibitors (MAPKi) by expressing cIAP2 (or LacZ as a control) in MeWo cells, which are *BRAF*^{WT} and express a low basal level of cIAP2 (Fig. 1A). Cells were treated with increasing concentrations of the MEKi, trametinib, and growth was assessed via IncuCyte assay. While cIAP2-overexpressing cells grew comparably to control cells in the absence of treatment,

they were significantly less sensitive to MEKi (Fig. 1B and 1C). cIAP2 has been shown to inhibit cell death by blocking the activation of effector caspases (28). Consistently, MeWo cIAP2-overexpressing cells showed less PARP cleavage (Fig. 1D) following MEK inhibition compared to LacZ control cells.

Next, we generated MeWo-Luc-mCherry cells that persist in the presence of MEKi treatment (known hereafter as MEKi tolerant cells, or MTC). MeWo MTC grew similarly to their untreated control in the presence of either trametinib or PD0325901 (PD'901), indicating cross tolerance to MEKi (Suppl. Fig. S1A and S1B). MTC expressed high levels of cIAP2 (Fig. 1E) and showed reduced PARP cleavage following MEKi treatment compared to the naive counterpart (Fig. 1F). These data suggest that cIAP2 overexpression is sufficient to confer tolerance to MAPK targeting agents in melanoma cells, likely through the reduction of apoptosis.

Similar to observations in A375 CRT *BRAF* mutant cells tolerant to BRAFi plus MEKi ((14) and Suppl. Fig. S1C), cIAP2 upregulation was associated with a dramatic reduction in SOX10 expression in MeWo MTC compared to the parental counterpart (Fig. 1E). SOX10 loss is associated with increased invasive and slow growing phenotypes (14), raising the possibility that DTPs may show similar features. We observed increased expression of the invasive markers fibronectin (FN1) and WNT5, and the cyclin-dependent kinase inhibitor, p21^{Cip1}, in the MeWo MTC as well as the *BRAF* mutant CRT cells (Fig. 1E and 1G). Thus, DTP cells, like SOX10-deficient cells, exhibit more invasive phenotypic properties compared to their parental counterparts.

cIAP2 is regulated by the AP-1 complex in SOX10-deficient cells

Given that cIAP2 upregulation was sufficient to mediate drug tolerance to targeted therapy in melanoma, we next sought to investigate the mechanisms regulating its expression. High levels of cIAP2 were associated with loss of SOX10 ((14), Fig. 1E and Suppl. Fig. S1C). Analysis of publicly available Assay for Transposase-Accessible Chromatin-sequencing (ATAC-seq) data (11) showed an increase in chromatin accessibility at the *BIRC3* promoter following SOX10 knockdown in melanoma cells (Suppl. Fig. S2A). Corroborating the specificity of SOX10 in regulating cIAP2, gain of function studies confirmed that SOX10 re-expression in SOX10 knockout cells reduced cIAP2 expression (Fig. 2A). However, analysis of two publicly available SOX10 Chromatin Immunoprecipitation-sequencing (ChIP-seq) datasets (23, 24) did not identify any SOX10 peaks at the *BIRC3* promoter in two different melanoma cell lines (Suppl. Fig. S2B). These data indicate that while SOX10 is involved in repressing cIAP2 expression, this is unlikely to occur through direct binding to the cIAP2/*BIRC3* promoter.

cIAP2 is regulated by transcriptional and post-translational mechanisms. Previous studies showed cIAP2 degradation by cIAP1 ubiquitination (29, 30). While cIAP1 knockdown increased cIAP2 expression in A375 and MeWo parental cells (Suppl. Fig. S2C), similar results were observed in the A375 and MeWo crSOX10 knockout lines (Suppl. Fig. S2D). Thus, cIAP2 upregulation via cIAP1 appears to be independent of SOX10 status. Next, we investigated whether cIAP2 is regulated at the transcript level. qPCR analysis showed a significant upregulation of cIAP2/*BIRC3* transcript levels in A375 and MeWo crSOX10

knockout cells compared to their parental counterparts (Fig. 2B). While data from other cancer types identified NF κ B (31, 32, 33, 34) as a transcriptional regulator of *cIAP2/BIRC3*, knockdown studies did not indicate a requirement for NF κ B in regulating *cIAP2/BIRC3* expression in MeWo MTC and A375 CRT SOX10-deficient cells (Suppl. Fig. S2E).

To identify other potential transcription factors (TFs) involved in *cIAP2* regulation, we used Harmonizome to select TFs predicted to bind to the *cIAP2/BIRC3* promoter. Next, using our previously published RNA-sequencing (RNA-seq) dataset (14), we determined which of the identified TFs were significantly differentially regulated in A375 and MeWo crSOX10 knockouts versus controls (Fig. 2C). SOX9 was the most upregulated TF across the SOX10 knockout models (Fig. 2C). Publicly available single-cell RNA-sequencing (scRNA-seq) data (Suppl. Fig. S2F) and Western blot analysis (Suppl. Fig. S2G-S2I) confirmed upregulation of SOX9 in cells expressing low levels of SOX10 and high levels of *cIAP2/BIRC3* (35) and Fig. 1E and Suppl. Fig. S1C). Furthermore, analysis of a publicly available SOX9 ChIP-seq dataset in human colorectal cancer (22) identified a SOX9 peak at the *BIRC3* promoter (Suppl. Fig. S2J). Despite these findings, targeting SOX9 in SOX10-deficient CRT cell lines did not affect *cIAP2* expression (Suppl. Fig. S2K). Thus, we ruled out the involvement of SOX9 in *cIAP2* regulation in SOX10-deficient melanoma cells.

Previous studies have indicated that low levels of SOX10 are associated with the invasive/mesenchymal cell state and a shift to the AP-1 regulated transcriptome (9, 11). Consistently, RNA-seq (14) analysis showed a significant upregulation of several AP-1 protein complex components, including JUN, FOSL2, JUND and MAFK, in SOX10 knockout cells (Fig. 2C). Analysis of publicly available ChIP-seq data from the Encyclopedia of DNA Elements (ENCODE) (20, 21) showed the presence of JUND and FOSL2 ChIP peaks at the *BIRC3* promoter in other cancer cell lines, embryonic stem cell lines, and liver tissue (Fig. 2D). Importantly, Western blot analysis showed upregulation of JUND and FOSL2 expression levels in SOX10 knockout cells, which was reversed by SOX10 re-expression (Fig. 2E). These findings led us to test for either a JUND or FOSL2 requirement for *cIAP2* modulation in SOX10 knockout cells. While FOSL2 knockdown did not affect *cIAP2* expression, siRNA-mediated downregulation of JUND reduced *cIAP2* expression in SOX10 knockout cells (Fig. 2F and 2G). These data suggest that the AP-1 component, JUND, is required for *cIAP2* expression in SOX10 knockout cells.

Birinapant induces cell death in SOX10-deficient cells

We previously demonstrated that high levels of *cIAP2* in SOX10-deficient melanoma cells confer vulnerability to IAP inhibitors, such as birinapant (14). To characterize mechanistically how birinapant affects SOX10-deficient cells, we performed Annexin V/PI staining to measure cell death. While parental cells showed no significant change in Annexin V/PI staining following birinapant treatment, we observed a significant increase in cell death in SOX10-deficient cells (Fig. 3A and 3B). These data suggest that birinapant selectively targets SOX10-deficient cells by inducing cell death. Consistently, treatment of MeWo crSOX10 knockout and SOX10-deficient CRT cells with birinapant induced cleavage of PARP and caspase-3 (Fig. 3C and 3D). Increased cleavage of gasdermin E (GSDME), a

marker for a form of inflammatory programmed cell death known as pyroptosis, was also observed in SOX10-deficient cells following treatment with birinapant (Fig. 3C and 3D).

Although DTP cells emerge in the presence MAPK targeting agents, treatment with MAPKi can still affect the ERK1/2 pathway and cell cycle proteins in these cells (Fig. 3E and Suppl. Fig. S3A-S3D). Counterintuitively, previous studies indicated that the addition of MAPKi elicited a protective effect on cell growth in CRT drug tolerant cells treated with birinapant (14), and we observe similar results in MeWo SOX10 knockout cells (Suppl. Fig. S3E). This prompted us to assess the effects of BRAFi plus MEKi on birinapant-induced cell death in SOX10-deficient cells. Consistently, the addition of BRAFi plus MEKi led to a reduction in the cleavage of PARP, caspase-3 and GSDME induced by birinapant (Fig. 3F). A similar reduction in birinapant-induced cell death was observed in 3D spheroid models in the presence of BRAFi plus MEKi (Fig. 3G and Suppl. Fig. S3F). These data confirm that BRAFi plus MEKi elicit a protective effect on birinapant-induced cell death and suggest that birinapant is more efficient in targeting SOX10-deficient cells *in vitro* when used as a single agent, likely through induction of cell death.

Continuous scheduling of birinapant delays the onset of acquired resistance in patient-derived xenografts.

Drug tolerant states emerge during the MRD phase following exposure to treatment and likely consist of a reservoir of cells that seed acquired resistance. Results from clinical trial data with BRAFi plus MEKi indicate that continuous treatment regimens are more effective than scheduling (36). Conversely, our *in vitro* data suggest that BRAFi plus MEKi elicit a protective effect in SOX10-deficient cells treated with birinapant. Based on this evidence, we explored different scheduling regimens of birinapant in combination with BRAFi plus MEKi to maximize the effect of birinapant, and to test whether we can target DTP cells during MRD and delay the onset of BRAFi plus MEKi resistance. We utilized melanoma PDXs since they represent improved models for tumor heterogeneity (8, 37). We selected a *BRAF* mutant PDX since preclinical (8) and clinical studies (38) have confirmed the ability of BRAFi plus MEKi to induce tumor regression and MRD in this melanoma subgroup. Specifically, we utilized the TJUMEL41 PDX, which has been previously shown to regress to the MRD phase upon BRAFi plus MEKi treatment (17), and shows heterogeneous expression of SOX10 (Suppl. Fig. S4A). Mice harboring TJUMEL41 PDX were initially treated for four weeks with BRAFi plus MEKi to reach MRD, and then assigned to four different arms: i) BRAFi plus MEKi administered continuously, ii) BRAFi plus MEKi and birinapant alternated in biweekly cycles (1 week BRAFi plus MEKi and 1 week birinapant, two doses), iii) BRAFi plus MEKi and birinapant were alternated in a weekly cycle (4 days BRAFi plus MEKi followed by 3 days of control chow with one dose of birinapant on the second day), and iv) BRAFi plus MEKi and birinapant administered continuously (Fig. 4A). Contrary to our *in vitro* studies but in agreement with clinical data, the continuous regimen of BRAFi plus MEKi and birinapant significantly delayed tumor growth and improved mouse survival compared to BRAFi plus MEKi alone, while scheduled dosing of birinapant and BRAFi plus MEKi resulted in no significant improvement (Fig. 4B and Suppl. Fig. S4B-S4D). These data suggest that combining birinapant with BRAFi plus

MEKi, during the MRD phase, improves mouse survival and delays the onset of resistance in a PDX model.

Discussion

Drug tolerant states are associated with MRD and likely prelude the onset of acquired resistance to targeted therapies and ICI (8, 9, 10). While the transcriptomic profiles of drug tolerant cell states have been characterized, the mechanisms that allow persister cells to survive in the presence of treatment remain poorly defined. Identifying targetable vulnerabilities in DTP cells is likely to improve treatment for Stage III and IV melanomas. In this study, we focused on the upregulation of cIAP2 in SOX10-deficient DTP cells as a mechanism of tolerance to MAPK targeting agents. Our findings showed that cIAP2 overexpression is sufficient to mediate drug tolerance to MEKi, likely by reducing the levels of cell death in cutaneous melanoma models. Furthermore, we show that JUND is required for the expression of cIAP2 in SOX10-deficient cells. Using a PDX model, we demonstrated that initiating treatment with the cIAP1/2 inhibitor, birinapant, in combination with BRAFi plus MEKi during the MRD phase delays the onset of resistance. Our studies underscore the importance of identifying and exploiting DTP cell vulnerabilities to target MRD and improve the duration of response to the current standard-of-care therapies in melanoma.

The SOX10-deficient invasive state is one of the four drug tolerant phenotypes present in the minimal residual disease of *BRAF* mutant PDX models following BRAFi plus MEKi treatment (8). A key need in the field is to optimize targeted therapy approaches for *BRAF* WT melanoma, since BRAFi are contraindicated for these patients. In a *BRAF* WT model, a SOX10-deficient population, which we characterized as being DTP cells, emerges following chronic exposure to MEKi. These data suggest that the loss of SOX10 might be a common mechanism of tolerance to targeted therapy, independent of *BRAF* mutation status. SOX10-deficient DTP melanoma cells express high levels of cIAP2, which is sufficient to mediate drug tolerance. Other studies suggest that the loss of SOX10 in melanoma confers resistance to BRAFi and MEKi through activation of TGF- β signaling and upregulation of EGFR (39).

Many cancers express high levels of cIAP2 and other IAPs. For example, all four IAP family members were commonly elevated in prostate cancers, and cIAP2 expression was elevated in human colon cancer cells (31, 40). High cIAP2 expression has also been linked to resistance to chemotherapies in lung cancer and TRAIL-induced apoptosis in prostate cancers, as well as decreased sensitivity to proteasome inhibitors in multiple myeloma (41, 42, 43). IAP proteins are known for their anti-apoptotic activity via the inhibition of caspases. However, they are also implicated in the regulation of NF κ B and MAPK signaling pathways (29, 44, 45), autophagy (46, 47), and the epithelial-to-mesenchymal transition (48). In our study, increased expression of cIAP2 in DTP cells induced tolerance to MAPK targeting agents, likely by decreasing the levels of cell death. Due to its potential other roles, a future direction will be to investigate the involvement of cIAP2 in ERK1/2 reactivation, regulation of the extracellular matrix, and the epithelial-to-mesenchymal transition.

Expression of cIAP2/*BIRC3* is known to be controlled by transcriptional mechanisms, namely through NF κ B activation (31, 32, 33, 34), and post-translational mechanisms,

including via cIAP1 (29, 30), USP11 and USP19 (49, 50). We show that cIAP2 upregulation in SOX10-deficient cell models occurs at the transcript level. This finding agrees with patient-derived cell line data showing a strong upregulation of cIAP2 transcript levels in SOX10-negative melanoma cells in scRNA-seq datasets (14). Previous studies suggested that low levels of SOX10 are associated with a shift to an AP-1 regulated transcriptome (9, 11). Despite this knowledge, how changes in the transcriptional landscape of the invasive/mesenchymal state affect plasticity and adaptive responses to MAPKi remain to be characterized. In the present studies, we demonstrated that JUND, a member of the AP-1 protein complex, is required for cIAP2 upregulation in SOX10-deficient cells. These data suggest a direct mechanistic link between the activation of the AP-1 complex and tolerance to MAPK targeting agents.

Early studies in melanoma indicated that targeting IAP proteins with birinapant in combination with TNF- α treatment inhibits the growth of melanoma cells (51). Here, we show that birinapant is selectively lethal for SOX10-deficient DTP cells. This provides a strategy to target the subset of DTPs that display an invasive cell state. Other studies have proposed combination therapies to target the distinct NCSC DTP state, including RXR agonist with BRAFi plus MEKi (8). A second approach to ablating the NCSC subpopulation and delaying relapse in PDX models using Focal Adhesion Kinase inhibitors has also been reported (52). While previous studies propose upfront combination therapy to target MRD, we demonstrated that DTP cells can be targeted during the MRD phase. Combining birinapant with BRAFi plus MEKi during the MRD phase improved mouse survival and increased targeted therapy efficacy in a PDX model. Conversely, our *in vitro* data indicated that BRAFi plus MEKi elicited a protective effect on SOX10-deficient cells treated with birinapant. These differences between the *in vitro* and *in vivo* studies could be attributed to the presence of additional drug tolerant cell states within the MRD state (8), some of which may not respond to birinapant treatment. Thus, moving forward it will be important to characterize the persister cell populations that emerge following treatment with birinapant in combination with BRAFi plus MEKi.

Data from Phase I/II and Phase II clinical trials for patients with solid tumors demonstrated that combining birinapant with chemotherapy ([NCT01188499](#)) or Pembrolizumab is safe and well tolerated (53). Nonetheless, in our study, we observed severe weight loss in two of the nine mice enrolled in the birinapant plus BRAFi plus MEKi combination treatment arm, in contrast with the death of just one mouse in the BRAFi plus MEKi arm. This might indicate increased toxicity with the continuous triple combination, and emphasizes the importance of investigating and optimizing drug scheduling regimens.

In addition to the inhibitory effects on apoptosis, cIAP1/2 can display immune modulatory functions via inhibition of the non-canonical NF κ B pathway in immune cells (44, 54, 55, 56). Previous studies suggest that IAP antagonists can promote B cell survival (54, 55, 56), activation of dendritic cells (54), and deliver a broad co-stimulatory signal to T cells (54, 57). Furthermore, our data indicate that birinapant induces GSDME cleavage in SOX10-deficient DTP cells. Induction of GSDME-dependent pyroptosis has been previously associated with increased intra-tumoral levels of active dendritic cells and T cells (58). Future investigation of the effects of cIAP1/2 inhibitors on pyroptosis may yield insights

into alterations in the tumor immune microenvironment and the possibility of testing the combination of birinapant with ICi (59).

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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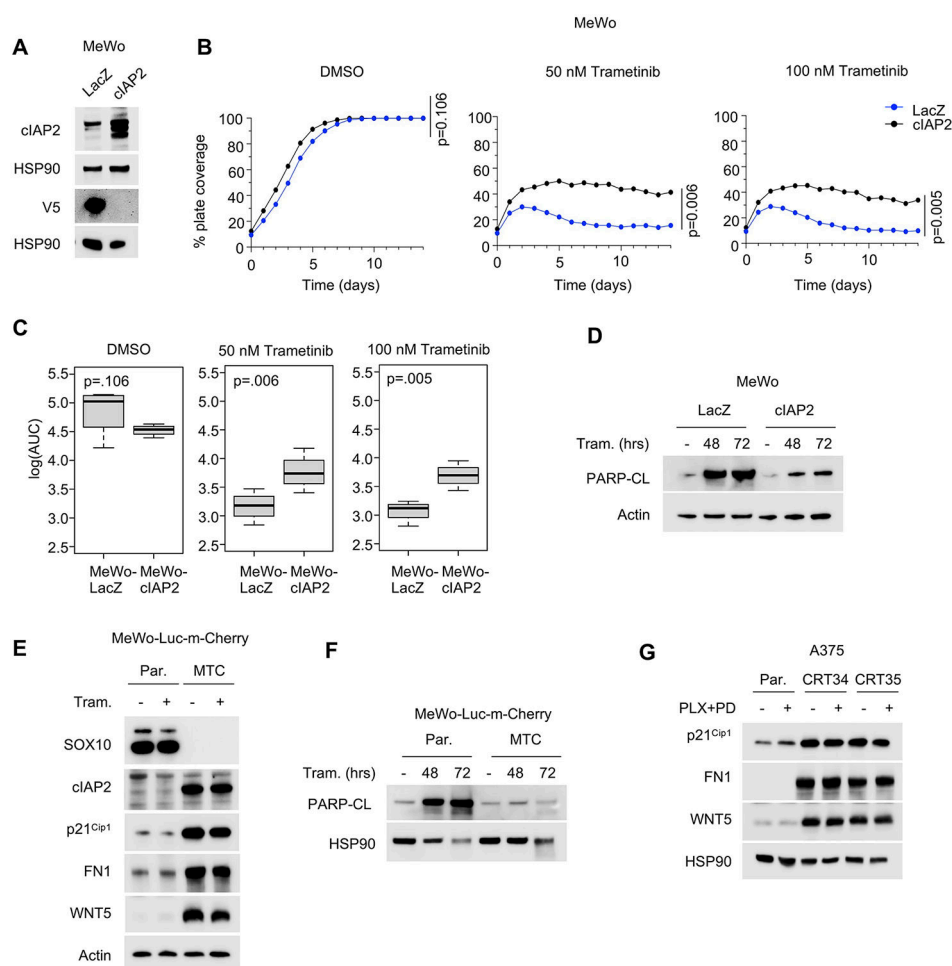


Figure 1. cIAP2 overexpression is sufficient to confer tolerance to MAPKi.

A. MeWo cells overexpressing LacZ control (MeWo-LacZ) or BIRC3 (cIAP2) (MeWo-cIAP2) were generated as described in Materials and Methods. The same number of MeWo-LacZ or MeWo-cIAP2 cells were seeded in six-well plates, lysed, and lysates Western blotted as indicated. **B.** MeWo-LacZ or MeWo-cIAP2 cells were plated, left untreated or treated with trametinib (50 or 100 nM), and imaged using an IncuCyte Live-Cell Analysis System. Treatment was renewed 3x per week for 14 days. Cell growth was determined as percent plate coverage. Shown is the mean from three independent experiments. *p*-value as indicated in the figure. **C.** Boxplot graphs of the log-transformed plate-specific Area Under Curve (AUC) for statistical analysis of data in 2B. Log-transformed plate-specific AUCs were analyzed in a two-way ANOVA model with the nominal effect of dose, cell line (MeWo-LacZ vs MeWo-cIAP2-overexpressing cells) and their interaction. The upper whisker boundary represents the largest data value within 1.5 inter-quartile range (IQR) above the third quartile, and the lower whisker boundary represents the smallest data value within 1.5 IQR below the first quartile. *p*-value as indicated in the figure. **D.** MeWo-LacZ or MeWo-cIAP2 cells were seeded in six-well plates, left untreated or treated with 50 nM trametinib for 48 or 72 hrs, lysed, and lysates Western blotted as indicated. **E.** MeWo MTC were generated as described in Materials and Methods from MeWo

pLentipuro/TO/V5/luciferase-mCherry cells (MeWo-Luc-m-Cherry). The same number of MeWo-Luc-m-Cherry or MeWo MTC were seeded in six-well plates, left untreated or treated with 50 nM trametinib for 24 hrs, lysed, and lysates Western blotted as indicated. **F.** The same number of MeWo-Luc-m-Cherry or MeWo MTC were seeded in six-well plates, left untreated or treated with 50 nM trametinib for 48 or 72 hrs, lysed, and lysates Western blotted as indicated. **G.** CRT34 and CRT35 were generated, as previously described (15). A375, CRT34, or CRT35 cells were seeded in six-well plates, left untreated or treated with PD'901 (35 nM) + PLX4720 (1 μ M) for 24 hrs, lysed, and lysates Western blotted as indicated.

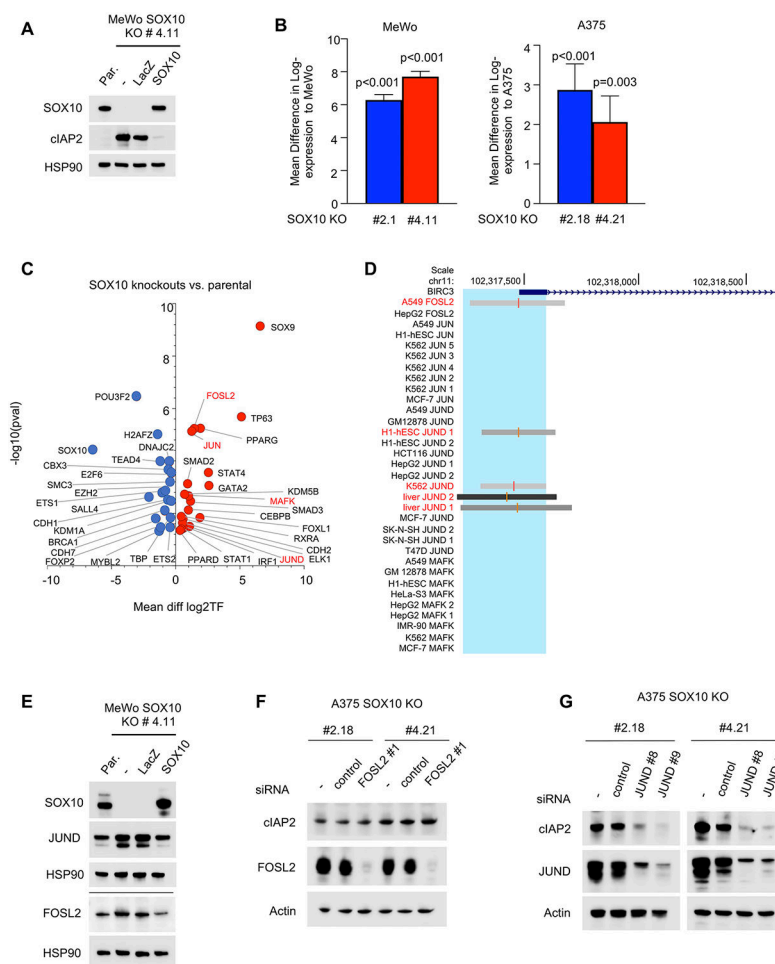


Figure 2. JUND is required for cIAP2 expression in SOX10-deficient cells.

A. MeWo crSOX10 #4.11 knockout cells were generated as previously described (14). MeWo crSOX10 #4.11-LacZ and MeWo crSOX10 #4.11-HA-SOX10 were generated as described in Materials and Methods. The same number of MeWo parental, MeWo crSOX10 #4.11, MeWo crSOX10 #4.11-LacZ and MeWo crSOX10 #4.11-HA-SOX10 were seeded in six-well plates, lysed, and lysates Western blotted as indicated. **B.** A375 crSOX10 #2.18, A375 crSOX10 #4.21, MeWo crSOX10 #2.1, and MeWo crSOX10 #4.11 knockout cells were generated as previously described (14). RNA was isolated from A375, A375 crSOX10 #2.18, A375 crSOX10 #4.21, MeWo, MeWo crSOX10 #2.1, and MeWo crSOX10 #4.11 cells. qRT-PCR was performed for cIAP2. Housekeeping gene GAPDH was used as an internal control. Average results from three independent experiments are shown. Error bars represent standard error of the mean (SEM). p -values as indicated in the figure. **C.** Volcano plot showing the transcription factors that are predicted to bind to the *BIRC3* promoter and are significantly differentially regulated across SOX10 knockout cells (MeWo crSOX10 #2.1 and #4.11 and A375 crSOX10 #2.18 and #4.21) compared to their parental counterparts. Harmonizome was used to identify transcription factors that are predicted to bind the *BIRC3* promoter. Statistical analysis was performed as described in Materials and Methods. AP-1 complex components are highlighted in red. **D.** AP-1 protein complex ChIP-seq data from

ENCODE indicate FOSL2 and JUND ChIP peaks at the *BIRC3* promoter region. **E.** The same number of MeWo parental, MeWo crSOX10 #4.11, MeWo crSOX10 #4.11-LacZ and MeWo crSOX10 #4.11-HA-SOX10 were seeded in six-well plates, lysed, and lysates Western blotted as indicated. **F.** The same number of A375 crSOX10 #2.18 or A735 crSOX10 #4.21 were seeded in six-well plates and treated with reagent alone, siGenome non-targeting control, or siFOSL2 for 48 hrs, lysed, and lysates Western blotted as indicated. **G.** The same number of A375 crSOX10 #2.18 or A735 crSOX10 #4.21 were seeded in six-well plates and treated with reagent alone, siGenome non-targeting control, or siJUND for 48 hrs, lysed, and lysates Western blotted as indicated.

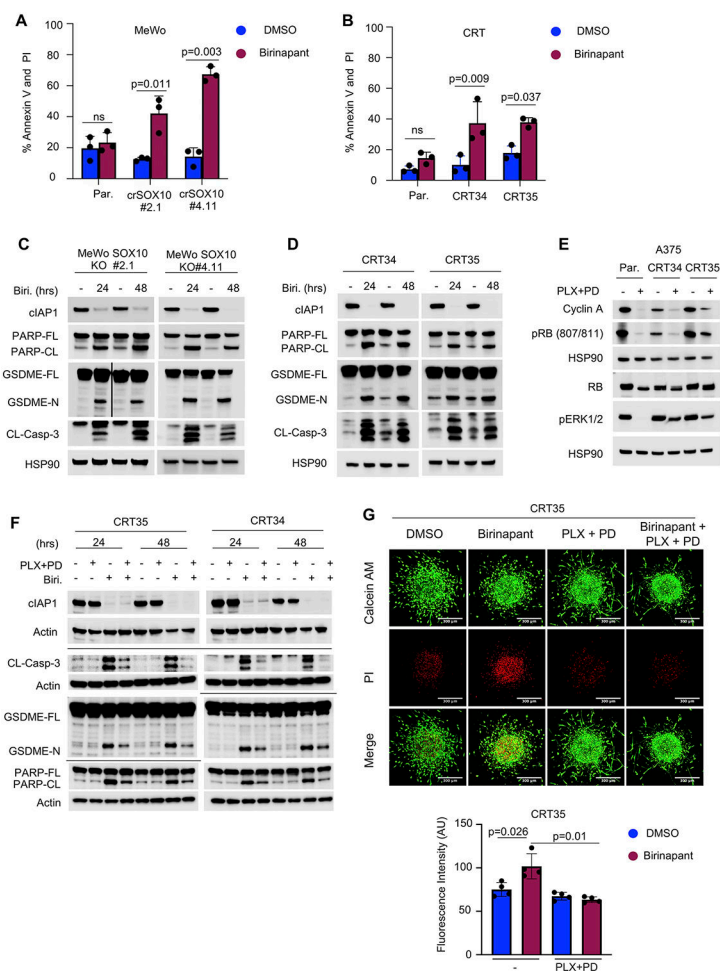


Figure 3. Birinapant induces cell death in SOX10-deficient cells *in vitro*

A. MeWo, MeWo crSOX10 #4.11 (3.0×10^5) or MeWo crSOX10 #2.1 (4.0×10^5) cells were plated and left untreated or treated with birinapant (100 nM). After 48 hrs, Annexin V/PI staining was performed as described in Materials and Methods. Shown is the average from three independent experiments. Percent of positive cells for Annexin V and PI staining was analyzed in a two-way ANOVA model. Error bars represent standard deviation. *p*-values as indicated in the figure. **B.** A375 (2×10^5) or CRT34 or CRT35 (3×10^5) cells were plated and left untreated or treated with birinapant (1 μ M). After 48 hrs, Annexin V/PI staining was performed as described in Materials and Methods. Shown is the average from three independent experiments. Percent Annexin V and PI staining was analyzed in a two-way ANOVA model. Error bars represent standard deviation. *p*-values as indicated in the figure. **C.** MeWo crSOX10 #4.11 (3×10^5) or MeWo crSOX10 #2.1 (4×10^5) cells were seeded in six-well plates, left untreated or treated with birinapant (100 nM) for 24 or 48 hrs, lysed, and lysates Western blotted as indicated. **D.** The same number of CRT34 or CRT35 cells were seeded in six-well plates, left untreated or treated with birinapant (1 μ M) for 24 or 48 hrs, lysed, and lysates Western blotted as indicated. **E.** The same number of A375, CRT34, or CRT35 cells were seeded in six-well plates, left untreated or treated with PD'901 (35 nM) + PLX4720 (1 μ M) for 24hrs, lysed, and lysates Western blotted as indicated. **F.** The same

number of CRT34 or CRT35 cells were seeded in six-well plates, left untreated or treated with birinapant (1 μ M) and/or PD'901 (35 nM) + PLX4720 (1 μ M) for 24 or 48 hrs, lysed, and lysates Western blotted as indicated. The black horizontal lines indicate the separation of blots from independent experiments. **G.** A375 CRT35 spheroids were treated for 4 days with 1 μ M birinapant and/or 1 μ M PLX470 + 35nM PD'901. Treatment was renewed every 48hrs. Spheroids were stained for live (Calcein AM) and dead (Propidium Iodide, PI) cells (upper panel). Representative images are shown. Scale bar, 300 μ M. Propidium Iodide fluorescence intensity was quantified using Image J (lower panel). Bar graphs show mean $\pm$ SD. AU = arbitrary units. Shown is the average of technical replicates. The experiment was repeated independently two times with similar results. *p*-values as indicated in the figure were calculated using Student's two-sample t-test assuming unequal variance.

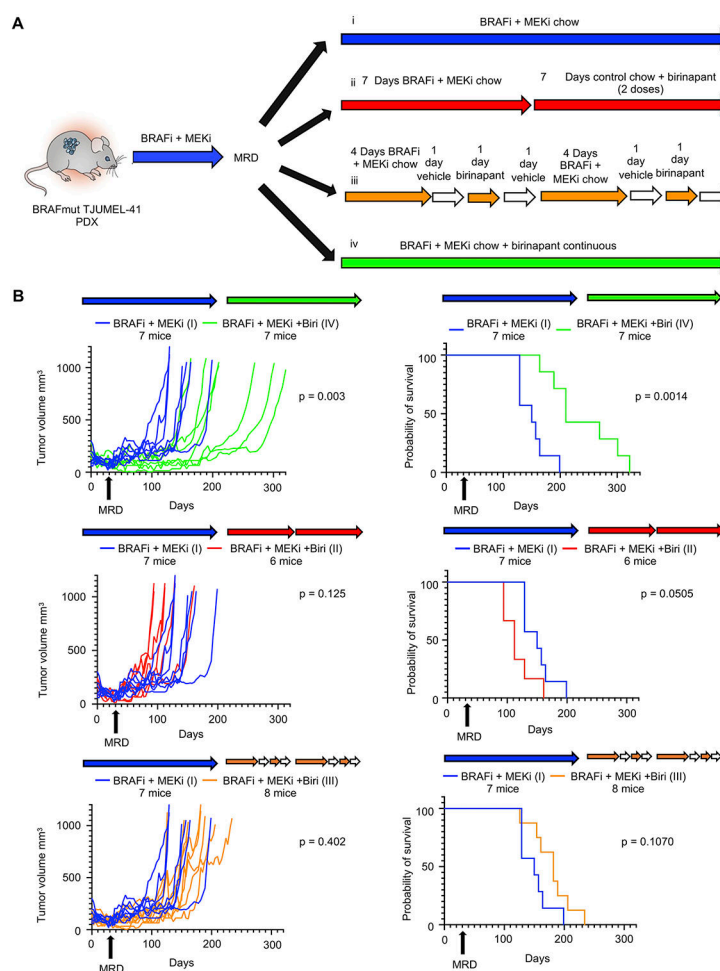


Figure 4. BRAFi plus MEKi in combination with birinapant prolongs mouse survival and delays tumor outgrowth *in vivo*.

A. Schematic illustrating BRAFi + MEKi + birinapant scheduling treatment arms in TJUMEL41 PDX mouse model. **B.** Tumor growth graphs comparing treatment arms of PDX TJUMEL41 and corresponding mouse survival curves, with number of mice per cohort as indicated. Details of administration of PLX4720 (200 PPM), PD'901 (7 PPM), and birinapant solution (3 mg/ml) are described in Materials and Methods. Significance was calculated as the time for the tumor volume to exceed 1000 mm³. *p*-values were calculated using a one-way ANOVA model with treatment group as a predictor. Post-hoc paired comparisons were performed with Tukey family-wise adjustment for multiple comparisons. Corresponding survival curves were analyzed using a Logrank (Mantel-Cox) test.